

Data Analysis for Business I (MG2008)

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Course Instructor(s)

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Final Exam

Total Time (Hrs): 3
Total Marks: 100
Total Questions: 7

25L-5575
Roll No

BSCFEJ2C
Section

Shahid
Student Signature

Important Instructions:

- Solve all parts of a question in the given sequence. (Compulsory; failure to follow the sequence may result in negative marking.)
- Clearly label each question number and part number (e.g., Q1(a), Q1(b), Q1(c)).
- Highlight or box the final answer of each part so that it is easy to identify during marking. Show complete calculations.
- Start every new question on a new page with proper label of question number at the top.
- Write all answers on the answer sheet only.

Do not write below this line

Attempt all the questions.

CLO # 1: Provide summaries using frequency distributions and graphical representation.

Q1: The following frequency distribution shows the monthly salaries (in \$1,000) of employees working in a software company during 2025. [12 marks]

Monthly Salary (\$000)	Frequency
20 up to 30	8
30 up to 40	15
40 up to 50	22
50 up to 60	30
60 up to 70	18
70 up to 80	7

✓ 20-29.5
✓ 30-39.5
✓ 40-49.5
50-59.5
60-69.5
70-79.5

100

Based on the above information, answer the following questions:

- a) What is the class interval? -10 or 9.5 if you add widths
- b) How many employees earn less than \$50,000 per month? $\rightarrow 22+8+15=45$
- c) What is the relative frequency of the 50 up to 60 class? $\rightarrow \frac{30}{100}=0.3$
- d) What is the midpoint of the 40 up to 50 class? $\rightarrow 45$
- e) The monthly salaries range between what two amounts? $80k$ and $20k$
 $\rightarrow \text{range} = 60k$
- f) What percentage of employees earn \$60,000 or more per month?
 $\rightarrow 55\%$

CLO # 2: Compute and interpret various measures of location and measures of variability.

Q2: A bank manager recorded the waiting time (in minutes) for every customer served during a particular morning shift. The waiting times are shown below: [10 Marks]
12, 18, 25, 15, 22, 30, 28, 17, 20, 35, 14, 26, 19

- a) Calculate mean waiting time.
- b) Calculate standard deviation.
- c) Calculate IQR.
- d) Calculate Coefficient of variation.

CLO # 3: Use basic counting techniques, laws of probability, specified discrete and continuous probability distributions in various business problems.Q3:

Q3: A trainer has a box containing 25 employee ID cards: 8 Marketing, 7 Finance, and 10 HR. One card is selected at random from the box. [9 Marks]

- a) If a Marketing card is selected and returned to the box, what is the probability that a Marketing card will be selected on the second draw?
- b) If a Marketing card is selected and not returned to the box, what is the probability that a Marketing card will be selected on the second draw?
- c) If an HR card is selected and not returned to the box, what is the probability that an HR card will be selected on the second draw?

CLO # 3: Use basic counting techniques, laws of probability, specified discrete and continuous probability distributions in various business problems.

Q4: At Allama Iqbal International Airport, passengers arrive at an average rate of 48 passengers per hour. Answer the following: [15 marks]

- a) Compute the probability of receiving three passengers in a 5-minute interval of time?
- b) Compute the probability of receiving exactly 10 passengers in 15-minutes?
- c) If the customer representative at the information desk, takes 5 minutes to complete the documentation of the last passenger, how many passengers do you expect to be waiting by that time? What is the probability that none will be waiting?
- d) If no passengers are currently being entertained, what is the probability that the customer representative can take a 3 minutes call for a meeting without being interrupted by any passenger?

CLO # 3: Use basic counting techniques, laws of probability, specified discrete and continuous probability distributions in various business problems.

Q5: A recent survey by a university health center found that the average daily screen time of students is 5.5 hours. Assume the standard deviation is 1.2 hours, and the screen time follows a normal distribution. [12 marks]

- a) What is the probability that a randomly selected student spends more than 7 hours per day on screens?
- b) What is the probability that a randomly selected student spends 4 hours or less per day on screens?
- c) Health experts recommend that students should limit screen time to between 4 and 6.5 hours per day. What percentage of students fall within this recommended range?

CLO # 3: Use basic counting techniques, laws of probability, specified discrete and continuous probability distributions in various business problems.

Q6: A survey was conducted among employees of a technology company to determine their preferred professional development activity. The results are shown below: **[12 marks]**

Gender	Technical Training	Leadership Training	Communication Skills	Data Analytics	Total
Male	30	25	15	20	90
Female	20	15	25	30	90
Total	50	40	40	50	180

If an employee is selected at random.

- Find the probability that the employee prefers Data Analytics.
- Find the probability that the employee is Female and prefers Communication Skills.
- Given that an employee prefers Technical Training, find the probability that the employee is Male.
- Given that an employee is Female, find the probability that she prefers Data Analytics.

CLO # 4: Determine the relationship, fitting a simple linear regression model, and assess the goodness of fit of regression model.

Q7: A researcher wants to study whether there is a relationship between study hours per week (X) and exam score (Y) of students. **[30 marks]**

Student	Study Hours (X)	Exam Score (Y)
1	2	52
2	3	50
3	4	61
4	5	64
5	6	68
6	7	78
7	8	77
8	9	88

- Develop the simple linear regression equation:
- Calculate:
 - SST
 - SSR
 - SSE
- Find the coefficient of determination (R^2) and interpret it.
- Find the correlation coefficient (r) and interpret the relationship.